NAME_____SCHOOL_____

MASAMBANJATI CLUSTER

2022 MSCE MOCK EXAMINATION

PHYSICS

Subject Number: M164/II

Date:_____

Time Allowed: 2 hour sessions

PAPER II

10:00 onwards

PRACTICAL

(40 marks)

Instructions

1. This paper contains two sections, A and B.
2. In section A, there are two descriptive questions on practical work to be answered in one hour. Marks will be given for accurate and orderly presentation of facts supported by relevant diagrams
3. In section B there are two practical questions to be answered in one hour
4. You should spend 30 minutes on each question in section A. The 30 minutes period allowed for each question include 3 minutes of tidy up apparatus and have it checked by the subject teacher.

Question Number	Tick if answered	Do not write in these columns		Marker's name
1				
2				
3				
4				

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SECTION A (20 MARKS)

1. With the aid of a well labelled diagram, describe an experiment that would be used to show that different metals expand differently when heated equally.

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(10 marks)

2. With the aid of a well labeled diagram, describe an experiment that could be used to determine the coefficient of static friction of a flat wooden plank.

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(10 marks)

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SECTION B (20 MARKS)

3. You are provided with the following materials: 1 meter nichrome wire, 1 ammeter, 1 voltmeter, 6 connecting wires and 1 cell.

   a. Connect the materials as shown in **figure 1**:

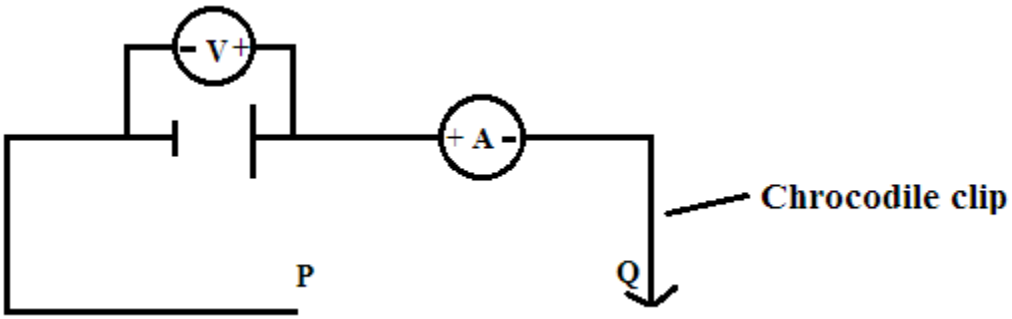

Figure 1

   b. Connect the nichrome wire on the gap PQ.

   c. Move the crocodile clip until the ammeter reads 0.9A

   d. Note and record the voltmeter reading in the table 1 below

Table 1

Current (A)	0.9	0.7	0.5	0.3
Voltage(V)				

(3 marks)

   b. Repeat steps b to c for the ammeter readings of 0.7A, 0.5A and 0.3A

   c. On the graph paper provided, plot a graph of voltage against current. **(5 marks)**

f. Use your graph to determine the emf of the cell.

(2 marks)

4. You are provided with the following materials:

1 meter rule retort stand, string (1m) and masses (50g, 100g, 150g, 200g and 250g) and stopwatch.

a. Set up the apparatus as shown in **figure 2.**

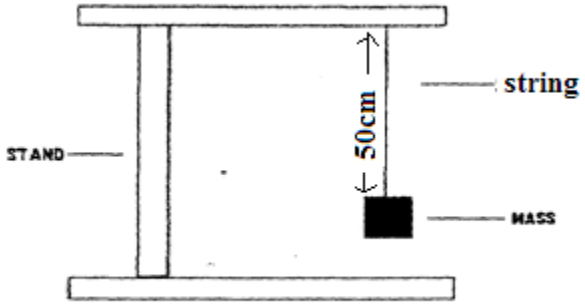

Figure 2

b. Hang 50g and let it oscillate freely.

c. Record the time taken for 10 oscillations.

d. Repeat step b to c without changing the length of the string for masses of 100g, 150g and 200g and 250g respectively.

(5 marks)

Mass (g)	Time(s) for 10 oscillations	Frequency (f)
50		
100		
150		
200		
250		

e. Complete the frequency column in the table.

f. Plot a graph of frequency against mass. **(4 marks)**

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f. Interpret the graph of mass and frequency of a pendulum.

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(1 mark)

END OF QUESTION PAPER

NB: This paper contains 7 printed pages.